

		any two relevant points, 1 each, max 2	B2	[2]
3	(a)	$pV/T = \text{constant}$	C1	
		$T = (6.5 \times 10^6 \times 30 \times 300)/(1.1 \times 10^5 \times 540)$	C1	
		$= 985 \text{ K}$	A1	[3]
		<i>(if uses °C, allow 1/3 marks for clear formula)</i>		
3	(b)	(i)		
		$\Delta U = q + w$		
		symbols identified correctly	M1	
		directions correct.....	A1	[2]
		(ii)		
		q is zero	B1	
		w is positive OR $\Delta U = w$ and U increases	B1	
		ΔU is rise in kinetic energy of <u>atoms</u>	M1	
		and mean kinetic energy $\propto T$	A1	[4]
		<i>(allow one of the last two marks if states 'U increases so T rises')</i>		